

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

1. Consider the statement: $n\sqrt{n}$ is $O(n \log(n))$. True or False?

Justify your answer using the definition of $O(\cdot)$.

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Answer:

1. The answer is False.

- (a) Let us start with the definition of $O(n \log(n))$:

$f(n)$ is $O(n \log(n))$ if there exists constants c and N , if for all $n > N$: $f(n) < cn \log(n)$.

- (b) We have that $\sqrt{n}$ grows faster than $\log(n)$, already for $N = 100$:

as $100^{1.5} = 1000$ and $100 \log_{10}(100) = 200$.

For convenience, we can use the $\log_{10}$ and use as constant to convert to $\log_2$.

- (c) The gap between $n\sqrt{n}$ and $n \log(n)$ increases as n increases, there is no constant c independent of n to relate the two functions.